

PROJECT SYNOPSIS

AIML 2nd Year – Project Synopsis Submission

SOLDIER HEALTH & FITNESS MONITORING SYSTEM

with AI-Powered Medical Note Intelligence & Soldier Morale Sentiment Tracking

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1. Introduction

Physical fitness and sound health are fundamental requirements for military personnel, since operational duties demand sustained endurance, strength, and continuous medical fitness. Presently, most defence units record medical checkups, injury history, and fitness scores through paper registers or disconnected spreadsheets, which makes it difficult to track long-term trends, spot early health risks, or gauge the psychological readiness of personnel. This synopsis proposes a centralized, web-based Soldier Health & Fitness Monitoring System, enhanced with two Artificial Intelligence / Natural Language Processing (NLP) capabilities adapted from the capstone problem statements “Smart Resume Screener” and “Brand Sentiment Tracker”, so that the system not only stores data but actively reads free-text medical notes and gauges soldier morale automatically.

2. Problem Statement

- Medical officers write checkup remarks and diagnoses as free-text notes, which are never structured or searched systematically, so early warning signs of deteriorating health are frequently missed.
- There is no centralized alert mechanism to flag soldiers who need urgent medical follow-up based on the content of their medical notes.
- Commanding officers have no reliable, data-driven way to gauge the mental well-being and morale of their unit, since psychological readiness is rarely tracked alongside physical fitness.
- Existing manual processes cannot scale to identify at-risk soldiers — physically or mentally — in units comprising several hundred personnel.

3. Objectives

- To design a centralized digital system for recording and managing soldiers' health, fitness, and medical data.
- To provide role-based access for soldiers, medical officers, commanding officers, and administrators.
- To apply NLP-based text parsing on medical officers' free-text checkup notes so that symptoms, conditions, and risk keywords are automatically extracted and used to rank soldiers by health-risk priority.
- To collect periodic soldier self-assessment feedback (well-being logs) and apply NLP sentiment classification to track morale trends over time, flagging soldiers with a declining sentiment pattern.

- To generate consolidated dashboards and reports that assist commanding officers in evaluating both the physical and psychological operational readiness of their unit.
- To reduce dependency on manual paperwork and minimise human error in health record management.

4. Existing System

Soldier health information is currently maintained through physical registers, individual medical files, and standalone spreadsheets prepared independently by different departments. Records are scattered, retrieval is slow, and no automated mechanism exists to flag soldiers who need follow-up care or who show declining morale. Free-text medical remarks are read manually, if at all, and are never mined for patterns.

5. Proposed System

The proposed system digitises and centralises soldier health and fitness management, and layers two AI-driven modules on top of the core platform, adapted directly from capstone problem statements 18 (Smart Resume Screener) and 20 (Brand Sentiment Tracker):

5.1 Intelligent Medical Note Parser & Risk Ranking Engine (*adapted from Project 18 – Smart Resume Screener*)

In the Smart Resume Screener, an NLP model parses uploaded resumes and ranks candidates against a job description by extracting context-aware skills. This project reuses the same underlying technique: when a medical officer uploads or types a checkup note, an NLP pipeline extracts symptoms, medical conditions, and risk-indicating keywords from the free text, and cross-references them against a defined set of medical risk criteria. Soldiers are then automatically ranked on a triage dashboard by health-risk priority, exactly as candidates are ranked against a job description, so medical officers can attend to the highest-risk cases first instead of reading every note manually.

5.2 Soldier Morale & Mental Wellness Sentiment Tracker (*adapted from Project 20 – Brand Sentiment Tracker*)

The Brand Sentiment Tracker classifies social media text as positive, neutral, or negative to gauge public perception over time. Here, soldiers periodically submit short self-assessment or well-being log entries through the portal. An NLP sentiment classification model scores each entry as positive, neutral, or negative, and the dashboard visualises morale trends over time for each soldier and for the unit as a whole — mirroring the way the Brand Sentiment Tracker visualises shifting public sentiment. A sustained negative trend automatically raises a notification for the commanding officer or medical officer to check in with the soldier, enabling early psychological intervention.

Additional improvements over the existing manual process:

- A single, centralized database storing soldier profiles, medical records, injury history, fitness results, and sentiment logs.
- Role-based dashboards presenting relevant information to soldiers, officers, and administrators.
- Automated calculation of fitness scores, health-risk ranking, and morale sentiment trends.
- Instant retrieval of any soldier's complete health, fitness, and well-being history through integrated search.

- Automated notifications for upcoming checkups, high health-risk flags, and declining morale trends.
- Report generation that assists commanding officers in assessing readiness — physical and psychological — at a glance.

6. Modules of the System

#	Module	Description
1	User Authentication	Secure, role-based login for soldiers, medical officers, commanding officers, and administrators.
2	Soldier Profile Management	Creation and maintenance of each soldier's core profile: personal details, service number, rank, and unit.
3	Health Monitoring	Recording of vital signs, BMI, and general health status from routine checkups.
4	Fitness Tracking	Logging and trend analysis of physical fitness test results (running time, push-ups, sit-ups, endurance).
5	Medical History	Chronological record of checkups, treatments, diagnoses, and injury history.
6	Intelligent Medical Note Parser (NLP)	Extracts symptoms/risk keywords from free-text medical notes and ranks soldiers by health-risk priority.
7	Morale Sentiment Tracker (NLP)	Classifies soldier well-being log entries as positive/neutral/negative and visualises morale trends over time.
8	Dashboard & Reports	Consolidated, role-specific summaries of health, fitness, and morale data.
9	Notifications	Automated alerts for pending checkups, high-risk medical flags, and declining morale trends.

7. Technology Stack

Layer	Technology
Frontend	HTML, CSS, Bootstrap, JavaScript
Backend	Python (Flask)
Database	MySQL
NLP / AI Modules	Python NLP libraries (e.g., spaCy / NLTK, scikit-learn) for keyword extraction and sentiment classification
Visualization	Chart.js / Flask-rendered dashboards for risk-ranking and sentiment trend graphs

8. Expected Outcome

- A fully functional web-based system for recording and managing soldiers' health, fitness, and morale data.

- Automated, NLP-driven triage ranking of soldiers based on medical note content, reducing manual review time.
- A morale sentiment dashboard that enables early detection of declining soldier well-being.
- Faster retrieval of individual soldier health history and improved unit-level readiness assessment for commanding officers.

9. Conclusion

By combining the centralized health and fitness record-keeping of the original system with NLP-driven medical note parsing (inspired by the Smart Resume Screener) and sentiment-based morale tracking (inspired by the Brand Sentiment Tracker), this enhanced Soldier Health & Fitness Monitoring System moves beyond simple data storage to proactive, AI-assisted decision support — helping medical officers prioritise care and helping commanding officers safeguard both the physical and psychological readiness of their units.